The finite J-norm is dominated by the discrete K-norm

(the direction $\mathcal{J} \lesssim \|\cdot\|_K$ of Janson's Lemma 1, over a monoid)

1 Setting and statement

Let α be an additive commutative monoid, M an additive commutative monoid with a preorder $\le$, and $x \mapsto |x|$ a map $\alpha \to M$ (`Modulus`) satisfying $|0| = 0$, $0 \le |x|$,

$$|x + y| \le |x| + |y| \qquad (x, y \in \alpha), \tag{1}$$

and the following *refined Riesz decomposition*:

(R) if $|x| \le |y| + |z|$, then there are $x_0, x_1 \in \alpha$ with

$$x = x_0 + x_1, \qquad |x_0| \le |y|, \quad |x_1| \le |z|, \qquad |x_0| \le |x|, \quad |x_1| \le |x|.$$

These four conditions are the fields `modulus_zero`, `modulus_nonneg`, `modulus_add_le` and `exists_decomp` of `Modulus.IsDecomposable`. The first two give $|0| \le |x|$ (`modulus_zero_le`). The instances are the following.

- On $\mathbb{R}$ and $\mathbb{C}$, $M = [0, \infty]$, $|x|$ is the norm, and (R) holds with $x_0 = cx$, $x_1 = (1 - c)x$, $c = \min(1, |y|/|x|)$ (`Modulus.isDecomposable_ofENorm`, for every real normed space).
- On $[0, \infty]$, $M = [0, \infty]$, $|x| = x$, $x_0 = \min(x, y)$ and $x_1 = x - x_0$.
- On products, M is the product of the value monoids, $|\cdot|$ is taken pointwise, and (R) is checked pointwise.

Let A_0, A_1 be quasinorms on α with values in $[0, \infty]$ that are *solid*: $|x| \le |y|$ implies $\|x\|_i \le \|y\|_i$, where $\|\cdot\|_i$ is the quasinorm of A_i. No other property of A_0, A_1 is used; in particular their quasi-triangle constants never appear.

Fix $0 < \theta < 1$ and $0 < q \le \infty$. For $\nu \in \mathbb{Z}$, $x, u \in \alpha$ put

$$K_\nu(x) = \inf_{x = x_0 + x_1} \big(\|x_0\|_0 + 2^\nu \|x_1\|_1\big), \qquad J_\nu(u) = \max\big(\|u\|_0,\ 2^\nu \|u\|_1\big),$$

and $\Delta = \{x \in \alpha : \|x\|_0 < \infty,\ \|x\|_1 < \infty\}$. For a family $f = (f_\nu)_{\nu \in S}$ in $[0, \infty]$ indexed by a countable set S, $\|f\|_{\ell^q}$ is $(\sum_{\nu \in S} f_\nu^q)^{1/q}$ if $q < \infty$ and $\sup_{\nu \in S} f_\nu$ if $q = \infty$. The two norms to be compared are

$$\|x\|_K = \big\|(2^{-\nu\theta} K_\nu(x))_{\nu \in \mathbb{Z}}\big\|_{\ell^q} \qquad (\texttt{discreteKMethod}),$$

$$\mathcal{J}(x) = \inf\Big\{ \big\|(2^{-\nu\theta} J_\nu(u_\nu))_{\nu \in \mathbb{Z}}\big\|_{\ell^q} : u \colon \mathbb{Z} \to \alpha \text{ finitely supported, } \textstyle\sum_{\nu \in \mathbb{Z}} u_\nu = x \Big\} \qquad (\texttt{jInfNorm}).$$

Since $J_\nu(0) = 0$, a family $u \colon \{-N, \dots, N-1\} \to \alpha$ with $\sum_{\nu=-N}^{N-1} u_\nu = x$, extended by zero to $\mathbb{Z}$, is admissible, with cost $\|(2^{-\nu\theta} J_\nu(u_\nu))_{\nu=-N}^{N-1}\|_{\ell^q}$. Conversely every admissible u is supported in $\{-N, \dots, N-1\}$ for some N. Hence $\mathcal{J}$ is equivalently the infimum of these costs over $N \in \mathbb{N}$ and $u \colon \{-N, \dots, N-1\} \to \alpha$.

Theorem 1. *For every $x \in \Delta$,*

$$\mathcal{J}(x) \le 2^{\min(\theta,\,1-\theta)}\,\|x\|_K.$$

The constant is at most $\sqrt{2}$ and does not depend on q or on the couple.

Idea of the proof. For each level μ choose a near-optimal decomposition $x = y_\mu + z_\mu$ for $K_\mu(x)$. Using (R), split x into $2N$ pieces, the piece u_ν being dominated by $y_{\nu+1}$ and by z_ν. Then $J_\nu(u_\nu) \lesssim K_{\nu+1}(x)$, except for one of the two norms of each of the two end pieces, which is controlled only by x itself. After weighting, these end terms decay geometrically in N, because $0 < \theta < 1$ and $x \in \Delta$. The ℓ^q-norm is estimated through its r-th power, $r = \min(q, 1)$, which is subadditive with constant 1. The error terms vanish in the limit, so the main term keeps the constant 2^θ for every q. Swapping A_0 and A_1 replaces θ by $1-\theta$ in both norms, so the bound with $2^{1-\theta}$ follows from the bound with 2^θ for the swapped couple.

2 Elementary facts

Lemma 2. *For every $x \in \alpha$ and $\nu \in \mathbb{Z}$, $K_\nu(x) \le K_{\nu+1}(x)$.*

Proof. For any decomposition $x = x_0 + x_1$, $\|x_0\|_0 + 2^\nu\|x_1\|_1 \le \|x_0\|_0 + 2^{\nu+1}\|x_1\|_1$. Taking the infimum over all decompositions gives the claim. □

The same argument shows that $t \mapsto \inf_{x=x_0+x_1}\big(\|x_0\|_0 + t\|x_1\|_1\big)$ is monotone on $[0,\infty]$ for every $x \in \alpha$; in Lean this is `monotone_kNorm`. The lemma is its case $2^\nu \le 2^{\nu+1}$ (`ENNReal.zpow_le_of_le`).

Lemma 3. *For every $x \in \alpha$, $\mu \in \mathbb{Z}$ and $\delta > 0$ there is a decomposition $x = y + z$ with $\|y\|_0 + 2^\mu\|z\|_1 \le K_\mu(x) + \delta$.*

Proof. If $K_\mu(x) = \infty$, any decomposition works, e.g. $x = 0 + x$. Otherwise $K_\mu(x) < K_\mu(x) + \delta$, and by the definition of the infimum some decomposition has cost below $K_\mu(x) + \delta$ (`exists_decomp_lt_of_lt_kNorm`). □

An additive error is used because the infimum need not be attained, and a multiplicative error $1 + \varepsilon$ is useless when $K_\mu(x) = 0$.

Lemma 4. *Put $r = \min(q, 1)$. For $f, g\colon \mathbb{Z} \to [0,\infty]$,*

$$\|f + g\|_{\ell^q}^r \le \|f\|_{\ell^q}^r + \|g\|_{\ell^q}^r.$$

Proof. If $q \ge 1$, including $q = \infty$, then $r = 1$ and this is Minkowski's inequality. Let $0 < q < 1$, so that $r = q$ and $\|f\|_{\ell^q}^q = \sum_{\nu\in\mathbb{Z}} f_\nu^q$. For $s, t \in [0,\infty]$ we have $(s+t)^q \le s^q + t^q$. This is clear if $s + t$ is 0 or ∞; otherwise it follows from $\lambda^q \ge \lambda$ for $\lambda \in [0,1]$, applied to $\lambda = s/(s+t)$ and $\lambda = t/(s+t)$. Summing this inequality with $s = f_\nu$, $t = g_\nu$ over $\nu \in \mathbb{Z}$ gives the claim. □

In Mathlib, Lemma 4 is `inhmgELpNorm_add_le`. For $q \ne 0$, $\|f\|_{\ell^q}^r$ equals `inhmgELpNorm f q Measure.count`, by

- `inhmgELpNorm_eq_eLpNorm` if $q \ge 1$,
- `inhmgELpNorm_eq_eLpNorm_pow` if $q \le 1$.

Lemma 5. *Let $f, g\colon \mathbb{Z} \to [0,\infty]$.*

(a) If $c \in [0,\infty)$ and $f_\nu \le c\, g_\nu$ for all $\nu \in \mathbb{Z}$, then $\|f\|_{\ell^q} \le c\|g\|_{\ell^q}$.

(b) For every bijection σ of $\mathbb{Z}$, $\|(f_{\sigma(\nu)})_{\nu\in\mathbb{Z}}\|_{\ell^q} = \|f\|_{\ell^q}$.

(c) If $S \subseteq \mathbb{Z}$ is finite with n elements and $c \in [0,\infty]$, then $\|c\,\mathbf{1}_S\|_{\ell^q} \le c\, n^{1/q}$, where $1/\infty = 0$.

Proof. (a) The ℓ^q-norm is monotone and positively homogeneous. (b) Sums and suprema are invariant under reindexing by a bijection. (c) If $q < \infty$, then $\|c\,\mathbf{1}_S\|_{\ell^q}^q = nc^q$. If $q = \infty$, then $\|c\,\mathbf{1}_S\|_{\ell^\infty}$ is c or 0, hence at most $c = c\,n^0$. □

In Lean:

(a) `eLpNorm_le_mul_eLpNorm_of_ae_le_mul'` with $c : \mathbb{R}_{\ge 0}$;

(b) `eLpNorm_comp_measurePreserving` with `Function.Bijective.measurePreserving_count`;

(c) `eLpNorm_indicator_const_le` with `Measure.count_apply_finset`.

The lemma `Function.Bijective.measurePreserving_count` (a measurable bijection preserves `Measure.count`) is not in Mathlib; it is added in this project. Every function on $\mathbb{Z}$ is measurable and a.e. strongly measurable, by `Measurable.of_discrete` and `AEStronglyMeasurable.of_discrete`.

3 Splitting x into $2N$ pieces

This is the only place where (R) is used. The first piece is split off r, and the rest of r is split by induction, so no subtraction is needed.

Proposition 6. *Let $x \in \alpha$, $a \in \mathbb{Z}$, $m \in \mathbb{N}$, put $I = \{a, \dots, a+m\}$, and for $\mu = a+1, \dots, a+m$ let $x = y_\mu + z_\mu$ be arbitrary decompositions. Let $r \in \alpha$ with $|r| \le |x|$. Then there is $u\colon \mathbb{Z} \to \alpha$ with $u_\nu = 0$ for $\nu \notin I$, $\sum_{\nu\in\mathbb{Z}} u_\nu = r$ and, for every $\nu \in I$,*

(a) $|u_\nu| \le |y_{\nu+1}|$ if $\nu < a+m$;

(b) $|u_\nu| \le |z_\nu|$ if $\nu > a$;

(c) $|u_\nu| \le |r|$.

Proof. By induction on m, for all a and r simultaneously.

The case $m = 0$. Let $u_a = r$ and $u_\nu = 0$ for $\nu \ne a$. Then (a) and (b) are vacuous, and (c) holds by reflexivity.

The step from m to $m+1$. Since $x = y_{a+1} + z_{a+1}$, inequality (1) and transitivity give

$$|r| \le |x| \le |y_{a+1}| + |z_{a+1}|.$$

Applying (R) to r gives $r = v + r'$ with

$$|v| \le |y_{a+1}|, \qquad |r'| \le |z_{a+1}|, \qquad |v| \le |r|, \qquad |r'| \le |r|. \tag{2}$$

By the last inequality, $|r'| \le |x|$, so the induction hypothesis applies to the start $a+1$, the length m and the remainder r'. It gives $u' \colon \mathbb{Z} \to \alpha$ vanishing outside $I' = \{a+1, \dots, a+1+m\}$ with $\sum_{\nu \in \mathbb{Z}} u'_\nu = r'$ and (a)–(c) relative to I' and r'. Define $u_a = v$ and $u_\nu = u'_\nu$ for $\nu \neq a$. Then u vanishes outside $I = \{a\} \cup I'$, and since $u'_a = 0$, $\sum_{\nu \in \mathbb{Z}} u_\nu = v + \sum_{\nu \in \mathbb{Z}} u'_\nu = v + r' = r$.

At $\nu = a$, (a) and (c) are the first and third inequality of (2), and (b) is vacuous. Let $\nu \in I'$, so $u_\nu = u'_\nu$. Property (a) is that of u', since the right end of I' is that of I. Property (c) follows from $|u'_\nu| \le |r'| \le |r|$. For (b), if $\nu > a+1$ it is that of u'. At $\nu = a+1$ the induction hypothesis gives no condition (b), because $a+1$ is the left end of I'; instead, (c) for u' and the second inequality of (2) give $|u'_{a+1}| \le |r'| \le |z_{a+1}|$. □

Both new conditions of (R) are needed. The condition $|x_1| \le |x|$ gives $|r'| \le |x|$, without which the induction hypothesis could not be applied; the condition $|x_0| \le |x|$, through (c), gives (b) at the left end of I'. For the same reason (c) is stated relative to r and not to x.

In Lean, Propositions 6 and 7 are `exists_decomp_Icc` and `exists_decomp_Ico_symm`, with (a)–(c) stated for every $\nu \in \mathbb{Z}$. For $\nu \notin I$ they hold because $u_\nu = 0$ and $|0| \le |w|$ for every $w \in \alpha$.

Proposition 7. *Let $N \ge 1$, $x \in \alpha$, and for $\mu = -N+1, \dots, N-1$ let $x = y_\mu + z_\mu$ be arbitrary decompositions. Then there is $u \colon \mathbb{Z} \to \alpha$ vanishing outside $\{-N, \dots, N-1\}$ with $\sum_{\nu \in \mathbb{Z}} u_\nu = x$ and, for every $\nu \in \{-N, \dots, N-1\}$,*

(a) $|u_\nu| \le |y_{\nu+1}|$ if $\nu \le N-2$;

(b) $|u_\nu| \le |z_\nu|$ if $\nu \ge -N+1$;

(c) $|u_\nu| \le |x|$.

Proof. Apply Proposition 6 with $a = -N$, $m = 2N-1$ and $r = x$; then $I = \{-N, \dots, N-1\}$. □

4 The pointwise estimate

Lemma 8. *Let $x \in \alpha$, $\delta > 0$ and $N \ge 1$. For $\mu = -N+1, \dots, N-1$ choose decompositions $x = y_\mu + z_\mu$ as in Lemma 3, i.e.*

$$\|y_\mu\|_0 + 2^\mu \|z_\mu\|_1 \le K_\mu(x) + \delta, \tag{3}$$

and let $u \colon \mathbb{Z} \to \alpha$ be given by Proposition 7. Put $E = 2^{-N}\|x\|_1 \, \mathbf{1}_{\{-N\}} + \|x\|_0 \, \mathbf{1}_{\{N-1\}}$. Then for every $\nu \in \mathbb{Z}$,

$$J_\nu(u_\nu) \le K_{\nu+1}(x) + \delta \, \mathbf{1}_{\{-N, \dots, N-1\}}(\nu) + E_\nu.$$

Proof. If $\nu \notin \{-N, \dots, N-1\}$, then $u_\nu = 0$, and $J_\nu(0) = 0$ because quasinorms vanish at 0. Let $\nu \in \{-N, \dots, N-1\}$, so that the indicator equals 1. We bound each of the two entries of $J_\nu(u_\nu) = \max\bigl(\|u_\nu\|_0, \; 2^\nu \|u_\nu\|_1\bigr)$ by one summand of the right-hand side; solidity turns each modulus inequality of Proposition 7 into an inequality of quasinorms.

The entry $\|u_\nu\|_0$. If $\nu \le N-2$, then by (a) and (3) at $\mu = \nu+1$, $\|u_\nu\|_0 \le \|y_{\nu+1}\|_0 \le K_{\nu+1}(x) + \delta$. If $\nu = N-1$, then by (c), $\|u_\nu\|_0 \le \|x\|_0$, which is the second summand of E_{N-1}.

The entry $2^\nu \|u_\nu\|_1$. If $\nu \ge -N+1$, then by (b), (3) at $\mu = \nu$, and Lemma 2, $2^\nu \|u_\nu\|_1 \le 2^\nu \|z_\nu\|_1 \le K_\nu(x) + \delta \le K_{\nu+1}(x) + \delta$. If $\nu = -N$, then by (c), $2^{-N}\|u_\nu\|_1 \le 2^{-N}\|x\|_1$, which is the first summand of E_{-N}. □

In Lean this is `jNorm_le_kNorm_add`, with the two indicator families of E written as `Pi.single`s.

5 The bound with 2^θ

Proposition 9. *For every $x \in \Delta$, $\mathcal{J}(x) \le 2^\theta \|x\|_K$.*

Fix $x \in \Delta$, and put $r = \min(q, 1)$ and $a_\mu = 2^{-\mu\theta} K_\mu(x)$ for $\mu \in \mathbb{Z}$, so that $\|a\|_{\ell^q} = \|x\|_K$. Fix $N \ge 1$ and $\delta > 0$, let u be as in Lemma 8, and put $b_\nu = 2^{-\nu\theta} J_\nu(u_\nu)$ for $\nu \in \mathbb{Z}$. The end terms are

$$e_{-N} = 2^{N\theta} 2^{-N} \|x\|_1 = 2^{-(1-\theta)N} \|x\|_1, \qquad e_{N-1} = 2^{-(N-1)\theta} \|x\|_0,$$

so that $2^{-\nu\theta} E_\nu = e_{-N} \mathbf{1}_{\{-N\}}(\nu) + e_{N-1} \mathbf{1}_{\{N-1\}}(\nu)$ for every $\nu \in \mathbb{Z}$. Since u vanishes outside $\{-N, \dots, N-1\}$ and $\sum_{\nu \in \mathbb{Z}} u_\nu = x$, the family u is admissible in the definition of $\mathcal{J}$, so $\mathcal{J}(x) \le \|b\|_{\ell^q}$.

Lemma 10. *For every $\nu \in \mathbb{Z}$,*

$$b_\nu \le 2^\theta a_{\nu+1} + \delta\, 2^{N\theta}\, \mathbf{1}_{\{-N,\dots,N-1\}}(\nu) + e_{-N} \mathbf{1}_{\{-N\}}(\nu) + e_{N-1} \mathbf{1}_{\{N-1\}}(\nu).$$

Proof. Multiply the inequality of Lemma 8 by $2^{-\nu\theta}$. Since $2^{-\nu\theta} = 2^\theta\, 2^{-(\nu+1)\theta}$, the first term becomes $2^\theta a_{\nu+1}$. The second term vanishes unless $\nu \ge -N$, and then $\theta > 0$ gives $2^{-\nu\theta} \le 2^{N\theta}$. The third term becomes $e_{-N} \mathbf{1}_{\{-N\}}(\nu) + e_{N-1} \mathbf{1}_{\{N-1\}}(\nu)$. □

The main term carries no indicator, so that Lemma 5(b) applies to it.

By Lemma 10 and Lemma 5(a) with $c = 1$, $\|b\|_{\ell^q}$ is at most the ℓ^q-norm of the sum of the four families

$$(2^\theta a_{\nu+1})_{\nu \in \mathbb{Z}}, \qquad \delta\, 2^{N\theta}\, \mathbf{1}_{\{-N,\dots,N-1\}}, \qquad e_{-N} \mathbf{1}_{\{-N\}}, \qquad e_{N-1} \mathbf{1}_{\{N-1\}}.$$

By Lemma 4, applied three times, $\|b\|_{\ell^q}^r$ is at most the sum of the r-th powers of their norms. By Lemma 5(a) and (b) with $\sigma(\nu) = \nu + 1$, the norm of the first family is at most $2^\theta \|a\|_{\ell^q} = 2^\theta \|x\|_K$. By Lemma 5(c), the norms of the other three are at most $\delta\, 2^{N\theta} (2N)^{1/q}$, e_{-N} and e_{N-1}. Hence

$$\mathcal{J}(x)^r \le 2^{\theta r} \|x\|_K^r + \big(\delta\, 2^{N\theta} (2N)^{1/q}\big)^r + e_{-N}^r + e_{N-1}^r. \tag{4}$$

The left-hand side of (4) depends on neither δ nor N, so δ may be chosen depending on N. For $N \ge 1$ put

$$\delta_N = 2^{-N\theta} (2N)^{-1/q} N^{-1},$$

which is positive because $2^{N\theta}(2N)^{1/q}$ is finite. Then $\delta_N\, 2^{N\theta} (2N)^{1/q} = N^{-1}$, and (4) with $\delta = \delta_N$ reads

$$\mathcal{J}(x)^r \le 2^{\theta r} \|x\|_K^r + N^{-r} + e_{-N}^r + e_{N-1}^r.$$

Now let $N \to \infty$. Since $r > 0$, $N^{-r} \to 0$. Since $\theta < 1$ and $\|x\|_1 < \infty$, $e_{-N} \to 0$; since $\theta > 0$ and $\|x\|_0 < \infty$, $e_{N-1} \to 0$. As $r > 0$, their r-th powers tend to 0 as well. Since addition on $[0, \infty]$ is continuous, this gives $\mathcal{J}(x)^r \le 2^{\theta r} \|x\|_K^r$, and taking r-th roots proves Proposition 9. □

No assumption $\|x\|_K < \infty$ is needed, and no case distinction between $q < \infty$ and $q = \infty$. In Lean, δ_N is not given by the formula above: `ENNReal.exists_pos_mul_lt` gives some $\delta_N > 0$ with $\delta_N\, 2^{N\theta} (2N)^{1/q} < N^{-1}$, which avoids the algebra in $[0, \infty]$, and `choose` turns these into a sequence. The limit is taken with `ge_of_tendsto` and `Filter.eventually_ge_atTop 1`, since (4) holds only for $N \ge 1$; the term N^{-r} tends to 0 by `ENNReal.tendsto_inv_nat_nhds_zero`.

6 Symmetry, and the bound with $2^{1-\theta}$

In this section the couple and θ are written as arguments. For $t \in [0,\infty]$ and $x, u \in \alpha$ put

$$K(t,x;A_0,A_1) = \inf_{x=x_0+x_1} \big(\|x_0\|_0 + t\|x_1\|_1\big), \qquad J(t,u;A_0,A_1) = \max\big(\|u\|_0,\ t\|u\|_1\big)$$

(`kNorm` and `jNorm`), so that $K_\nu(x) = K(2^\nu, x; A_0, A_1)$ and $J_\nu(u) = J(2^\nu, u; A_0, A_1)$. We write $K_\nu(x; A_0, A_1)$, $J_\nu(u; A_0, A_1)$, $\|x\|_K(A_0, A_1; \theta)$ and $\mathcal{J}(x; A_0, A_1; \theta)$ for the quantities of Section 1. The swapped couple (A_1, A_0) is `A.swap` (`Couple.swap`). For $g\colon \mathbb{Z} \to [0,\infty]$ put $\Phi_\theta(g) = \|(2^{-\nu\theta} g_\nu)_{\nu\in\mathbb{Z}}\|_{\ell^q}$ (`discretePhiFunctional`). Then $\|x\|_K(A_0, A_1; \theta) = \Phi_\theta(\nu \mapsto K_\nu(x; A_0, A_1))$, and the cost of an admissible family u in $\mathcal{J}(x; A_0, A_1; \theta)$ is $\Phi_\theta(\nu \mapsto J_\nu(u_\nu; A_0, A_1))$.

Lemma 11. *Let $0 < t < \infty$ and $x, u \in \alpha$. Then*

(a) $K(t,x;A_1,A_0) = t\,K(t^{-1},x;A_0,A_1)$;

(b) $J(t,u;A_1,A_0) = t\,J(t^{-1},u;A_0,A_1)$.

In particular, for $\nu \in \mathbb{Z}$,

$$K_\nu(x;A_1,A_0) = 2^\nu K_{-\nu}(x;A_0,A_1), \qquad J_\nu(u;A_1,A_0) = 2^\nu J_{-\nu}(u;A_0,A_1).$$

Proof. Since $t \neq 0, \infty$, multiplication by t commutes with infima and with max on $[0,\infty]$, and $t\,t^{-1} = 1$. Hence $t\,K(t^{-1},x;A_0,A_1)$ is the infimum of $t\|x_0\|_0 + \|x_1\|_1$ over all decompositions $x = x_0 + x_1$, and $t\,J(t^{-1},u;A_0,A_1) = \max(t\|u\|_0, \|u\|_1)$. For (a), reindex the infimum by $(x_0,x_1) \mapsto (x_1,x_0)$; since α is commutative, this map is a bijection of the set of decompositions of x. (b) is the commutativity of max. The special cases are $t = 2^\nu$, where $(2^\nu)^{-1} = 2^{-\nu}$. □

In Lean (`Basic.lean`):

- (a) is `kNorm_swap`. Its proof uses `ENNReal.mul_iInf_of_ne`, `ENNReal.mul_inv_cancel`, `Equiv.iInf_congr` with `Equiv.prodComm`, and `iInf_congr_Prop` with `add_comm`;
- (b) is `jNorm_swap`, by `mul_max`, `ENNReal.mul_inv_cancel` and `max_comm`;
- the dyadic cases are `kNorm_swap_zpow` and `jNorm_swap_zpow`, by `ENNReal.zpow_pos`, `ENNReal.zpow_ne_top` and `ENNReal.zpow_neg`.

Lemma 12. *For $g\colon \mathbb{Z} \to [0,\infty]$ and $\theta \in \mathbb{R}$, $\Phi_\theta(\nu \mapsto 2^\nu g_{-\nu}) = \Phi_{1-\theta}(g)$.*

Proof. Since $2^{-\nu\theta}2^\nu = 2^{\nu(1-\theta)} = 2^{-(-\nu)(1-\theta)}$, the family defining the left-hand side is $(h_{-\nu})_{\nu\in\mathbb{Z}}$, where $h_\mu = 2^{-\mu(1-\theta)} g_\mu$ is the family defining $\Phi_{1-\theta}(g)$. Apply Lemma 5(b) with $\sigma(\nu) = -\nu$. □

In Lean this is `discretePhiFunctional_reflect` (`KMethod.lean`). The weight in `discretePhiFunctional` is the real power $2^{(-\nu)\cdot\theta}$, while 2^ν is an integer power. The identity of the weights follows from `ENNReal.rpow_intCast`, `ENNReal.rpow_add` (with `two_ne_zero` and `ENNReal.ofNat_ne_top`) and `push_cast; ring` on the exponents. The reflection is `eLpNorm_comp_measurePreserving` with `AEStronglyMeasurable.of_discrete` and `neg_bijective.measurePreserving_count .of_discrete`.

Proposition 13. *For $x \in \alpha$ and $\theta \in \mathbb{R}$,*

(a) $\|x\|_K(A_1, A_0; \theta) = \|x\|_K(A_0, A_1; 1-\theta)$;

(b) $\mathcal{J}(x; A_1, A_0; \theta) = \mathcal{J}(x; A_0, A_1; 1-\theta)$.

Proof. (a) By Lemma 11 and Lemma 12 with $g_\mu = K_\mu(x; A_0, A_1)$,

$$\|x\|_K(A_1, A_0; \theta) = \Phi_\theta\big(\nu \mapsto 2^\nu K_{-\nu}(x; A_0, A_1)\big) = \Phi_{1-\theta}(g) = \|x\|_K(A_0, A_1; 1-\theta).$$

(b) For $u\colon \mathbb{Z} \to \alpha$ let $w\colon \mathbb{Z} \to \alpha$, $w_\nu = u_{-\nu}$. By Lemma 11, $J_\nu(u_\nu; A_1, A_0) = 2^\nu J_{-\nu}(w_{-\nu}; A_0, A_1)$. So by Lemma 12 with $g_\mu = J_\mu(w_\mu; A_0, A_1)$, the cost of u for $(A_1, A_0; \theta)$ is $\Phi_{1-\theta}(g)$, the cost of w for $(A_0, A_1; 1-\theta)$. The family u is admissible if and only if w is: the support of w is the negative of the support of u, and $\sum_\nu w_\nu = \sum_\nu u_\nu$ because $\nu \mapsto -\nu$ is a bijection of $\mathbb{Z}$. Every family $\mathbb{Z} \to \alpha$ is of the form w for some u, namely for $u_\nu = w_{-\nu}$. Hence, as u runs through the admissible families, w runs through all admissible families, and the two infima in (b) are infima of the same set of costs. □

In Lean:

- (a) is `enorm_discreteKMethod_swap`. It is stated for the values of the extended norms, not as an equality of `EQuasinorm`s, since the constant field `C` of `discreteKMethod` need not be invariant under the swap. The variant for `A.swap` is `Couple.enorm_discreteKMethod_swap`.
- (b) is `jInfNorm_swap`, by `Function.Surjective.iInf_congr` along $u \mapsto u \circ$ `Neg.neg`, which is surjective by `neg_injective.surjective_comp_right`. The two Prop-indexed infima are handled by `iInf_congr_Prop`, using `Function.support_comp_eq_preimage`, `Set.neg_preimage` and `Set.finite_neg` for the support, and `finsum_comp_equiv` with `Equiv.neg` $\mathbb{Z}$ for the sum.

Corollary 14. *For every $x \in \Delta$, $\mathcal{J}(x) \le 2^{1-\theta}\|x\|_K$.*

Proof. The hypotheses are invariant under swapping A_0 and A_1 and replacing θ by $1-\theta$: (R) concerns α alone, both quasinorms are solid, $0 < 1-\theta < 1$, and the definition of Δ is symmetric in A_0 and A_1. So Proposition 9 applies to (A_1, A_0) and $1-\theta$. Together with Proposition 13(b) and (a), each taken with $1-\theta$ in place of θ, it gives

$$\mathcal{J}(x; A_0, A_1; \theta) = \mathcal{J}(x; A_1, A_0; 1-\theta) \le 2^{1-\theta}\|x\|_K(A_1, A_0; 1-\theta) = 2^{1-\theta}\|x\|_K(A_0, A_1; \theta). \quad \square$$

In Lean:

- Proposition 9 is `jInfNorm_le_two_rpow_mul_discreteKMethod`, and Theorem 1 is `jInfNorm_le_discreteKMethod`.
- Δ is `A.inf.finiteLocus`, defined through $J(1, \cdot\,; A_0, A_1)$. Its invariance under the swap is `finiteLocus_inf_comm`, which follows from `enorm_inf_comm`, that is, Lemma 11(b) with $t = 1$.
- `Couple.swap` is an `abbrev`, so instance search finds `A.swap.fst.IsSolid` from `A.snd.IsSolid`.
- $1 - (1-\theta) = \theta$ is `sub_sub_cancel`.

Proof of Theorem 1. Since $2^{\min(\theta, 1-\theta)}$ is 2^θ or $2^{1-\theta}$ (`min_choice`), the theorem follows from Proposition 9 and Corollary 14. □

7 Remarks

Remark 15 (Where the hypotheses are used).

- (R), (1) and transitivity of $\leq$: only in Proposition 6.
- $|0| = 0$ and $0 \leq |x|$: only in the Lean form of Proposition 6, through $|0| \leq |w|$ for $\nu \notin I$.
- Solidity of A_0, A_1: only in Lemma 8.
- $x \in \Delta$ and $0 < \theta < 1$: in $e_{-N}, e_{N-1} \to 0$. Each end uses one half of each condition: e_{-N} uses $\theta < 1$ and $\|x\|_1 < \infty$, e_{N-1} uses $\theta > 0$ and $\|x\|_0 < \infty$. Besides, $\theta > 0$ gives the bound $2^{-\nu\theta} \leq 2^{N\theta}$ in Lemma 10, where any finite bound would do.
- The quasi-triangle inequalities of A_0 and A_1: never. For ℓ^q, only Lemma 4 is used. Its constant is 1 for every q, which is why the constant of Theorem 1 does not depend on q; the quasi-triangle inequality of ℓ^q would contribute a factor $2^{1/q-1}$ for $q < 1$.

For $q = 0$, Mathlib's `eLpNorm` is identically 0. The family with $u_0 = x$ and $u_\nu = 0$ for $\nu \neq 0$ (`Pi.single 0 x`) is admissible, so $\mathcal{J}(x) = 0$ and the inequality is trivial. This case is excluded from the proof above because there $r = 0$.

Remark 16 (Why $x \in \Delta$ is necessary). Let $q \neq 0$. If $\mathcal{J}(x) < \infty$, some admissible ℓ^q-norm is finite, so each $J_\nu(u_\nu)$ is finite and each u_ν lies in Δ. Since Δ is an additive submonoid (by the quasi-triangle inequality with finite constants), the finite sum x lies in Δ. Hence $\mathcal{J} = \infty$ outside Δ, whereas $\|x\|_K$ may be finite there.